



C3 Photosynthesis

C · Interaction and interdependence · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Photosynthesis converts light energy into chemical energy stored in organic molecules. In the light-dependent reactions (thylakoid membrane), photosystems absorb light and use the energy to split water (photolysis), releasing oxygen, and to generate ATP (via chemiosmosis) and NADPH. In the light-independent reactions (Calvin cycle, stroma), the enzyme rubisco fixes CO₂ onto RuBP, and ATP and NADPH from the light-dependent reactions are used to reduce the fixed carbon to G3P, which regenerates RuBP and can be used to build glucose. The overall rate of photosynthesis is limited by whichever factor — light intensity, CO₂ concentration, or temperature — is furthest from optimal at a given time.

Worked example: follow the reasoning

A student measures the rate of photosynthesis of an aquatic plant at increasing light intensity, first at a low CO₂ concentration (0.02%) and then at a higher CO₂ concentration (0.1%), keeping temperature constant. At low CO₂, rate rises with light intensity but plateaus early at a low maximum rate. At high CO₂, rate rises further before plateauing at a higher maximum rate. Step 1: at low light intensity in both curves, light is limiting, since rate increases as light increases and both curves overlap. Step 2: at higher light intensity, the low-CO₂ curve plateaus while the high-CO₂ curve continues rising, showing CO₂ was limiting the low-CO₂ curve at that point (not light). Step 3: eventually the high-CO₂ curve also plateaus at a higher rate, at which point another factor (e.g. temperature or enzyme/rubisco capacity) becomes limiting. Conclusion: whichever factor is in shortest supply relative to the others at a given light intensity limits the overall rate of photosynthesis.

Quick check · multiple choice

In the light-dependent reactions, what immediately replaces the electrons lost from photosystem II?

- A Electrons from NADPH
- B Electrons from the splitting of water (photolysis)
- C Electrons generated in the Calvin cycle
- D Electrons released from CO₂ fixation

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 A student investigated the effect of CO₂ concentration on the rate of photosynthesis of a plant at high, constant light intensity and constant temperature (25°C). Rate increased steadily as CO₂ concentration rose from 0.01% to 0.06%, then remained constant as CO₂ concentration was increased further to 0.10%. [6]
 - (a) Identify the limiting factor for photosynthesis between 0.01% and 0.06% CO₂, and explain your reasoning. [2]
 - (b) Suggest, with a reason, what factor is most likely limiting the rate above 0.06% CO₂. [2]
 - (c) Explain, in terms of the Calvin cycle enzymes, why the rate cannot increase indefinitely simply by adding more CO₂. [2]

Final answer: _____



Answers and explanation

Question 1 [6]

- (a) CO₂ concentration is the limiting factor between 0.01% and 0.06%, because increasing CO₂ while light and temperature remain constant directly increases the rate of photosynthesis, showing CO₂ availability was restricting the rate.
- (b) Above 0.06%, another factor — most likely light intensity or temperature (both held constant in this experiment) — becomes limiting, since increasing CO₂ further no longer increases rate.
- (c) CO₂ is fixed onto RuBP by the enzyme rubisco in the Calvin cycle; once all rubisco (and other Calvin cycle enzyme) molecules are working at their maximum rate, they become saturated and cannot process CO₂ any faster, so adding more CO₂ beyond this point cannot increase the rate of carbon fixation.

Quick check answer

B. Photolysis splits water into electrons (replacing those lost from photosystem II), protons, and oxygen gas, which is released as a by-product.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Biology HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.